

Lab2 results document:

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Names of all students you discussed this lab with:

Section 1:

Output from script in item 2:

$y =$

1 2 3 3 2 1

$ny =$

-2 -1 0 1 2 3

Question 2(c): What does $[y, ny] = \text{myconv}(x, nx, h, nh)$ return? Why do you expect this to happen?

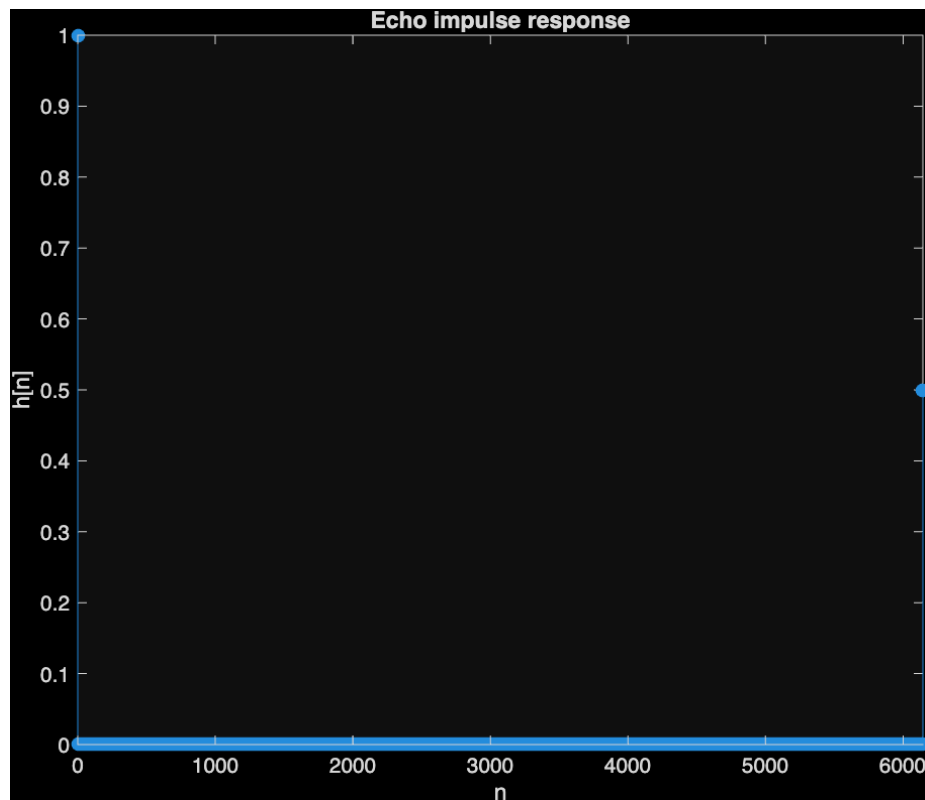
The result is expected because, for an LTI system, convolving the input signal with the system's impulse response produces exactly the same output as passing that input through the system, and in our case the input is an impulse, so the output is the same signal h

Question 2(d): What relationship do you expect between $\text{myconv}(x, nx, h, nh)$ and $\text{myconv}(h, nh, x, nx)$?

Convolution is commutative so both function calls have same output

Section 2:

Include your figure for item 2(b) below.



Question 2(e): What happens if you increase or decrease the parameters D and α ?

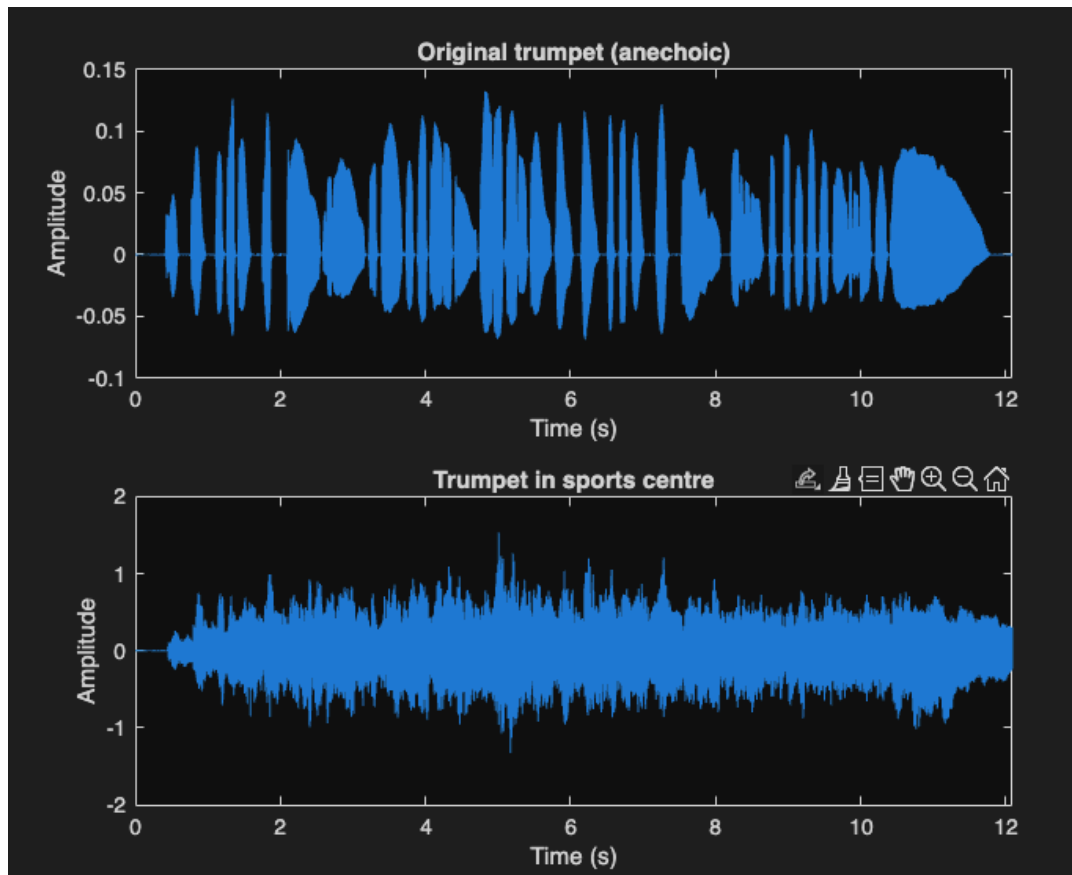
- Increasing D delays the echo more, decreasing D makes the echo closer
- Increasing α makes the echo louder, decreasing α makes the echo quieter.

Question 2(f): How many echoes do you hear?

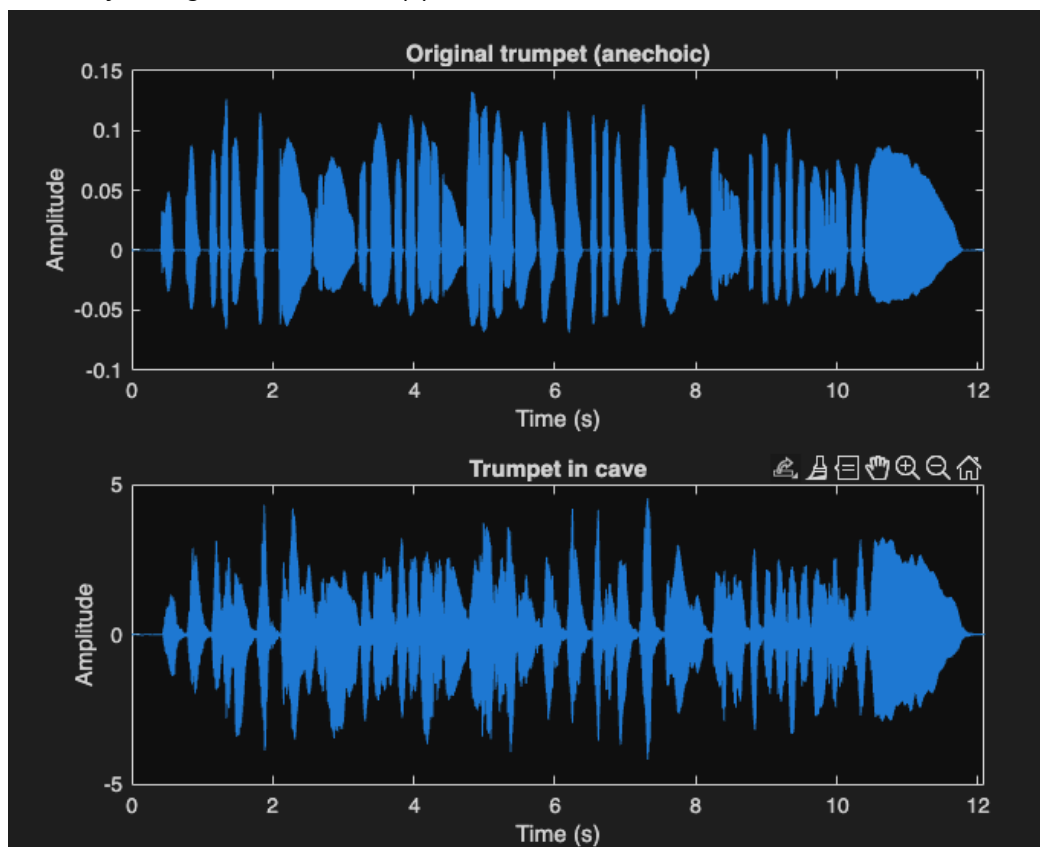
I can hear 2 echoes

Section 3:

Include your figure from item 2(c) below:

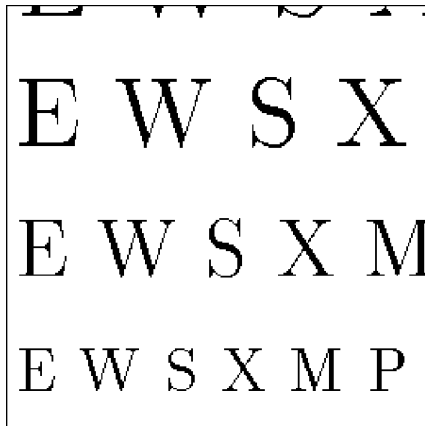


Include your figure from item 3(c) below:

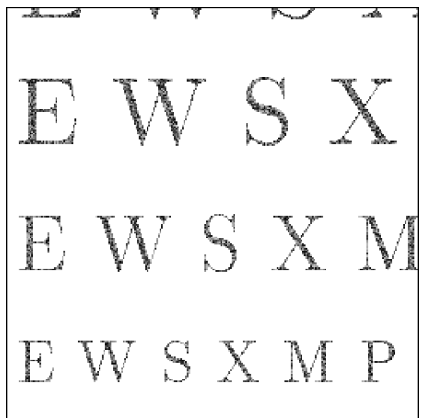


Section 4

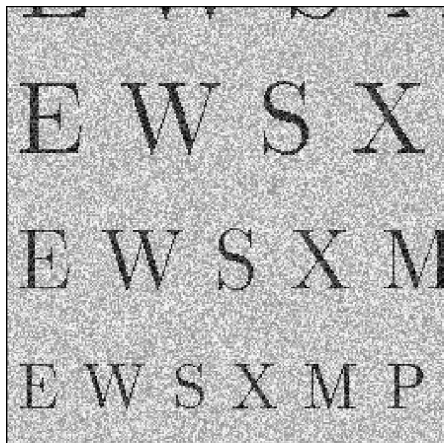
Include your image from item 2 below:



Include your image from item 5 below:



Include your image from item 6 below:



Question 3: How many pixels wide and how many pixels high is the image?

Height: 257

Width: 256

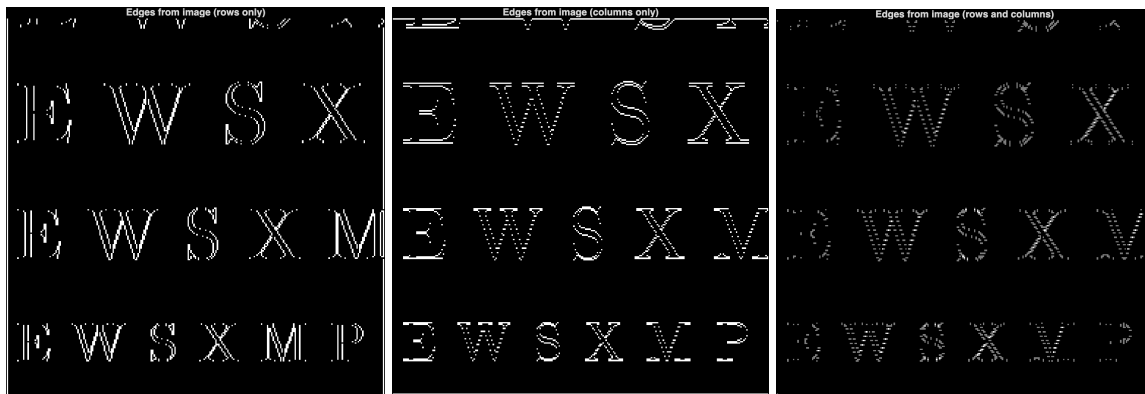
Question 7: What is the difference between the displayed images in items 5 and 6?

In task 5 the noisy image looks very bright and washed out, with many areas turning pure white because pixel values above 1 are clipped.

In task 6 the same noisy image has normal contrast again because it is rescaled before display; details of the letter are clearer but with visible grainy noise instead of big white patches.

Section 5:

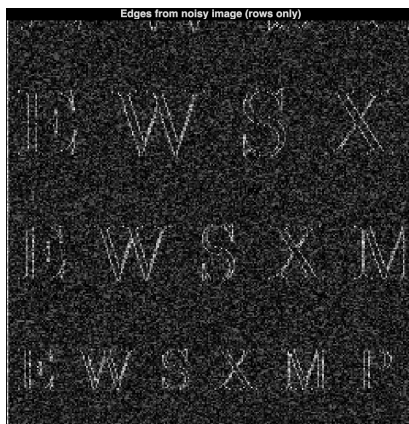
Include the images from item 1(d) below



Question 1(e): Based on the images you made in part (d), briefly explain how convolution of the rows and/or columns of the image with h , followed by an absolute value operation, enhances the edges of an image.

Convolution computes differences between neighbouring pixels (which are large at edges) and then makes all large differences bright, so object boundaries stand out while flat regions stay dark.

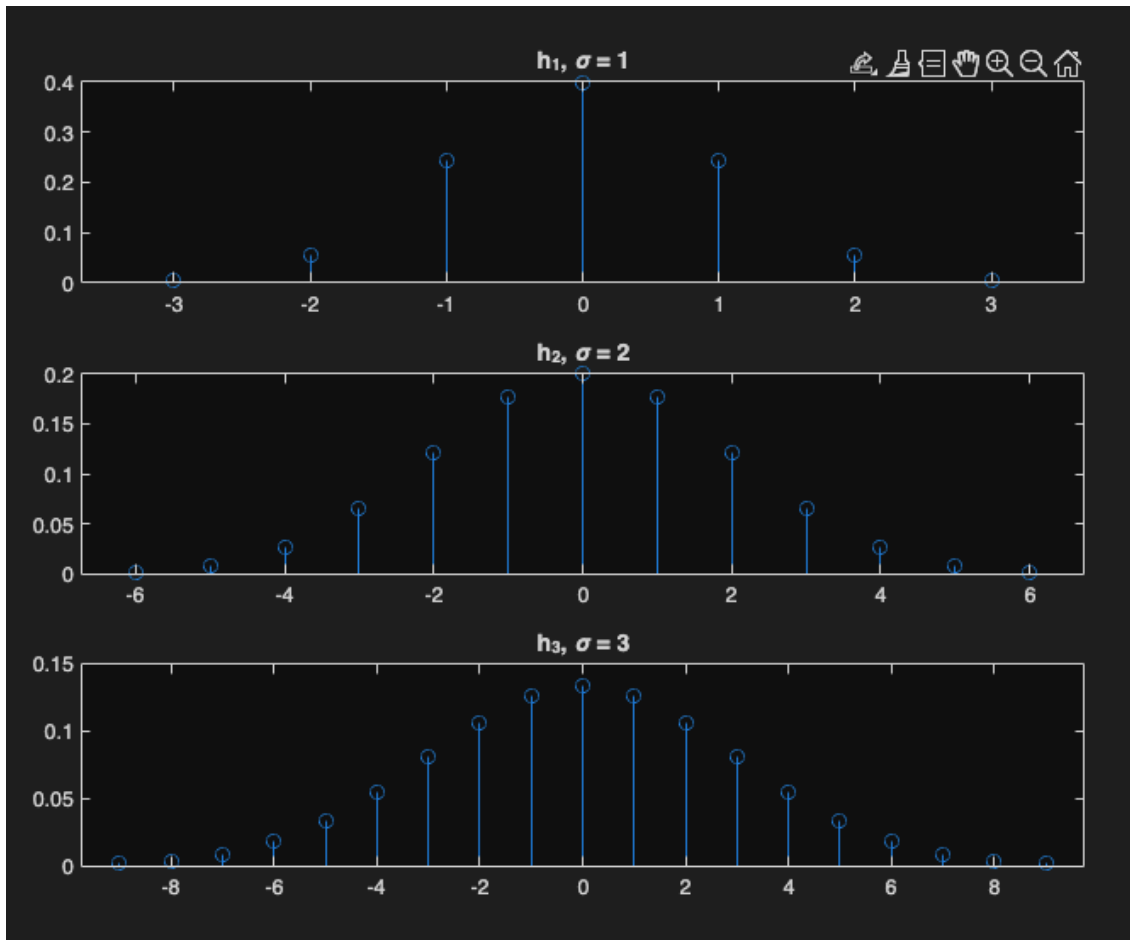
Include the image from item 2(b) below



Question 2(c): How does noise affect the result of the edge enhancement operation?

Noise introduces many small random intensity changes between neighbouring pixels, so the edge filter also highlights these fluctuations, making the edge image more speckled and less clean, with real edges harder to distinguish from noise.

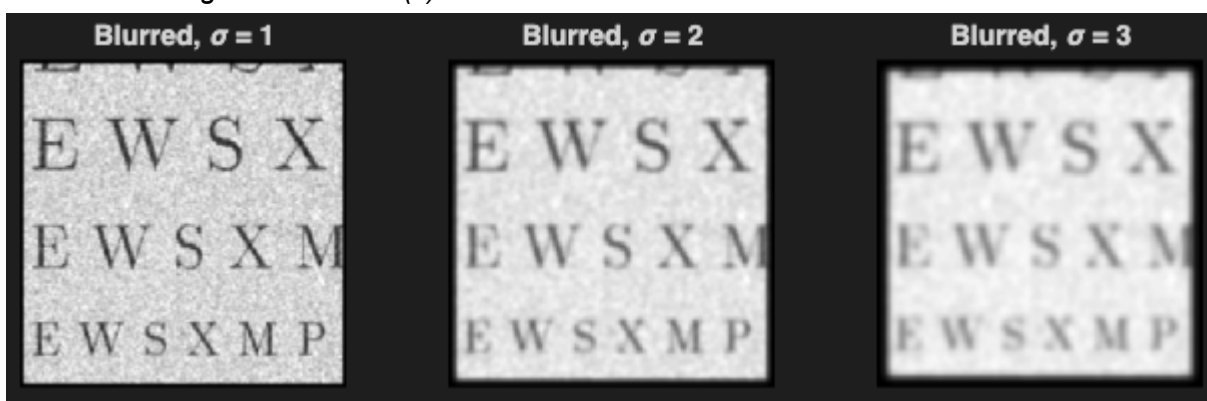
Include your plots from item 3(b) below:



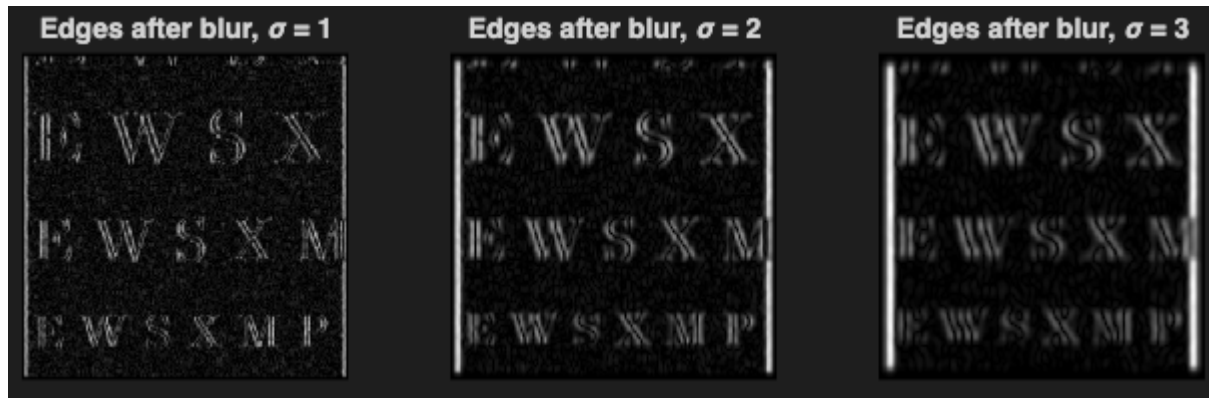
Question 3(b): What is the effect of changing σ ?

Increasing σ (variance) makes the Gaussian wider and flatter, spreading the weights over more samples and giving a stronger blur.

Include the images from item 3(c) below



Include the images from item 3(d) below



Question 3(e): How does first applying a Gaussian blur (with different values of σ) affect the performance of subsequent edge enhancement filtering?

Gaussian blurring with increasing σ smooths away small noisy fluctuations before edge detection, so real edges become cleaner and less speckled, but very large σ also softens and thickens the edges, reducing fine detail.

Code for section 1:***Code from section 1 item 1:***

Paste your code for myconv in here.

```
function [y, ny] = myconv(x, nx, h, nh)
    ax = nx(1);
    bx = nx(end);
    ah = nh(1);
    bh = nh(end);

    ay = ax + ah;
    by = bx + bh;
    ny = ay:by;

    Lx = length(x);
    Lh = length(h);
    Ly = Lx + Lh - 1;

    y = zeros(1, Ly);
    for n = 1:Ly
        for k = 1:Lx
            m = n - k + 1;
            if m >= 1 && m <= Lh
                y(n) = y(n) + x(k) * h(m);
            end
        end
    end
end
```

Code from section 1 item 2:

Paste your script in here.

```
[x, nx] = dtimpulse(0, 0, 0);

h = [1:1:3, 3:-1:1];
nh = -2:3;

[y, ny] = myconv(x, nx, h, nh)
[y, ny] = myconv(h, nh, x, nx)

x = [1 1 1];
nx = 2:4;

[y1, ny1] = myconv(x,nx,h,nh)
[y2, ny2] = myconv(h,nh,x,nx)
```

Code for section 2

Code from section 2 item 1:

Paste your echoIR function code here.

```
function [h, nh] = echoIR(D, alpha)
    if ~isscalar(alpha) || alpha <= 0 || alpha >= 1
        error('alpha must be a scalar strictly between 0 and 1.');
```

end

```
    nh = 0:D;
    h = zeros(1, D+1);
    h(1) = 1;
    h(end) = alpha;
end
```

Code from section 2 item 2:

Paste your script in here.

```
D = 6144;
alpha = 0.5;
[h, nh] = echoIR(D, alpha);
figure;
stem(nh, h, 'filled');
xlabel('n');
ylabel('h[n]');
title('Echo impulse response');
xlim([0 D]);
load handel;
ny = 0:(length(y)-1);
[yecho, nyecho] = myconv(y, ny, h, nh);
soundsc(yecho, Fs);
[yecho2, nyecho2] = myconv(yecho, nyecho, h, nh);
soundsc(yecho2, Fs);
```

Code for section 3

Code from section 3.1:

Paste your script in here.

```
[trumpet_y, trumpet_Fs] = audioread('trumpet.wav');
soundsc(trumpet_y, trumpet_Fs);
L_trumpet = length(trumpet_y)
[hSports, Fs_sports] = audioread('sportscentre.wav');
Fs_sports, trumpet_Fs
trumpetSports = conv(trumpet_y, hSports);
save('trumpetSports.mat', 'trumpetSports');
soundsc(trumpetSports, trumpet_Fs);
N_trumpet = length(trumpet_y);
N_trumpetSports = length(trumpetSports);
T_trumpet = (0:N_trumpet-1)/trumpet_Fs;
t_trumpetSports = (0:N_trumpetSports-1)/trumpet_Fs;
figure;
subplot(2,1,1);
plot(t_trumpet, trumpet_y);
xlabel('Time (s)');
ylabel('Amplitude');
title('Original trumpet (anechoic)');
xlim([0, t_trumpet(end)]);
subplot(2,1,2);
plot(t_trumpetSports, trumpetSports);
xlabel('Time (s)');
ylabel('Amplitude');
title('Trumpet in sports centre');
xlim([0, t_trumpet(end)]);
[hCave, Fs_cave] = audioread('cavemono.wav');
Fs_cave, trumpet_Fs
trumpetCave = conv(trumpet_y, hCave);
soundsc(trumpetCave, trumpet_Fs);
N_trumpetCave = length(trumpetCave);
t_trumpetCave = (0:N_trumpetCave-1)/trumpet_Fs;
figure;
subplot(2,1,1);
plot(t_trumpet, trumpet_y);
xlabel('Time (s)');
ylabel('Amplitude');
title('Original trumpet (anechoic)');
xlim([0, t_trumpet(end)]);
subplot(2,1,2);
plot(t_trumpetCave, trumpetCave);
xlabel('Time (s)');
ylabel('Amplitude');
title('Trumpet in cave');
xlim([0, t_trumpet(end)]);
```

Code for section 4:*Code from section 4.2:*

Paste your script in here.

```
load echart;  
imshow(echart);  
[p, q] = size(echart)  
echartnoisy = echart + 0.8*rand(size(echart));  
imshow(echartnoisy);  
imshow(echartnoisy, []);
```

Code for section 5:

Code from section 5.1:

Paste your script in here.

```
load echart;
echartnoisy = echart + 0.8*rand(size(echart));
h = [-1 1];
Yrow = conv2(1, h, echart);
Ycol = conv2(h', 1, echart);
Yboth = conv2(h', 1, conv2(1, h, echart));
%figure;
%imshow(echart);
figure;
imshow(abs(Yrow), []);
title('Edges from image (rows only)');
figure;
imshow(abs(Ycol), []);
title('Edges from image (columns only)');
figure;
imshow(abs(Yboth), []);
title('Edges from image (rows and columns)');
Zrow = conv2(1, h, echartnoisy);
figure;
imshow(abs(Zrow), []);
title('Edges from noisy image (rows only)');
sigma1 = 1;
sigma2 = 2;
sigma3 = 3;
n1 = -3*sigma1 : 3*sigma1;
n2 = -3*sigma2 : 3*sigma2;
n3 = -3*sigma3 : 3*sigma3;
h1 = (1/(sqrt(2*pi)*sigma1)) * exp(-n1.^2./(2*sigma1^2));
h2 = (1/(sqrt(2*pi)*sigma2)) * exp(-n2.^2./(2*sigma2^2));
h3 = (1/(sqrt(2*pi)*sigma3)) * exp(-n3.^2./(2*sigma3^2));
figure;
subplot(3,1,1); stem(n1, h1); title('h_1, \sigma = 1');
subplot(3,1,2); stem(n2, h2); title('h_2, \sigma = 2');
subplot(3,1,3); stem(n3, h3); title('h_3, \sigma = 3');
Zblur1 = conv2(h1, h1, echartnoisy);
Zblur2 = conv2(h2, h2, echartnoisy);
Zblur3 = conv2(h3, h3, echartnoisy);
figure;
subplot(1,3,1); imshow(Zblur1, []); title('Blurred, \sigma = 1');
subplot(1,3,2); imshow(Zblur2, []); title('Blurred, \sigma = 2');
subplot(1,3,3); imshow(Zblur3, []); title('Blurred, \sigma = 3');
Zedge1 = conv2(1, h, Zblur1);
Zedge2 = conv2(1, h, Zblur2);
Zedge3 = conv2(1, h, Zblur3);
```

```
figure;  
subplot(1,3,1); imshow(abs(Zedge1),[]); title('Edges after blur,  
\sigma = 1');  
subplot(1,3,2); imshow(abs(Zedge2),[]); title('Edges after blur,  
\sigma = 2');  
subplot(1,3,3); imshow(abs(Zedge3),[]); title('Edges after blur,  
\sigma = 3');
```